Econ 814 Recitation 11 Notes

Frank Gentile

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1 Types of models

There are 4 possible combinations of models studied in Econ 812:

- Deterministic Finite
- Deterministic Infinite
- Stochastic Finite
- Stochastic Infinite

For each of these combinations you can solve via Lagrangian or Dynamic Programming:

Deterministic

1.1 Finite

Lagrangian

$$\mathscr{L} = \sum_{t=0}^{T} \beta^t [u(c_t) + \lambda_t(-c_t - k_{t+1} + k_t)] + \beta^T \mu_T [k_{T+1} - 0]$$

With TVC: $\beta^T \mu_T k_{T+1} = 0$
DP

$$V_T(k_T) = \max u(c_T) + \beta V_{T+1}(k_{T+1})$$

With $k_{T+1} = 0$ so $V_T(k_T) = u(c_T) \implies c_T^* = k_T$

1.2 Infinite

Lagrangian

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t [-c_t - k_{t+1} + k_t]]$$

With TVC: $\lim_{t\to\infty} \beta^t \lambda_t k_{t+1} = 0$

DP

$$V(k) = \max_{c_t, k_{t+1}} u(c_t) + \beta V(k_{t+1})$$

And follow the steps below.

Stochastic

1.3 Finite

Lagrangian

$$\mathscr{L} = \sum_{t=0}^{T} \beta^t \sum_{s^t} \Pi(s^t)[u(c(s^t)) + \lambda(s^t)[-c(s^t) - k_{t+1} + z(s^t)k_t]] + \beta^T \sum_{s^t} \Pi(s^t)\mu_T(s^T)k(s^{T+1})$$

With TVC: $E_T[\beta^T \mu_T k_{T+1}] = 0$

DP

$$V(k(s^T), z(s^T)) = \max_{c(s^T), k(s^{T+1})} u(c(s^T)) + E_T[\beta V(k(s^{T+1}), z(s^{T+1}))]$$

With $V_{T+1} = 0$

1.4 Infinite

Lagrangian

$$\mathscr{L} = \sum_{t=0}^{T} \beta^t \sum_{s^t} \Pi(s^t)[u(c(s^t)) + \lambda(s^t)[-c(s^t) - k_{t+1} + z(s^t)k_t]]$$

With TVC: $\lim_{j\to\infty} E_t[\beta^{t+j} \lambda_{t+j} K_{t+j+1}] = 0$

DP

$$V(k(s^t), z(s^t)) = \max_{c(s^t), k(s^{t+1})} u(c(s^t)) + E_t[\beta V(k(s^{t+1}), z(s^{t+1}))]$$

And follow the steps below.

In general, the strategy to solving these models is to

- Set up either Lagrangian or Bellman
- Solve for Euler Equation
- Solve for policy functions

- Log linearize
- Solve for eigenvalues and eigenvectors to construct matrices for stacked state and control variables
- Solve for unstable and stable conditions in linear difference equations

2 Dynamic Programs

2.1 A simple example

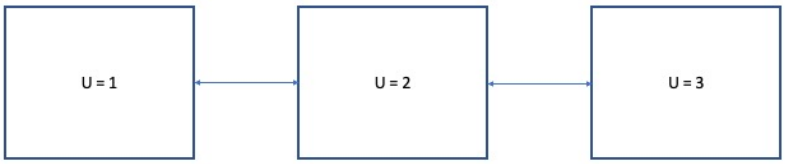


Figure 1: Three states of the world with utility equalling 1, 2, and 3

Imagine a world with three states (Figure 1) and two periods. Say, we are dropped into this world in the middle at U=2 at T=1 and can only move (freely) once before T=2.

What we are trying to accomplish is to maximize the value of our states. But we need some guiding light on how to make the optimal choice of our action (moving once).

This is where the policy function comes in. It is the policy that our actions follow to give the optimal answer. Now, there could be many policy functions. But we set up the problem to have a unique solution.

Some examples of policy functions might be "always left" or "always right" or "don't move" or "move right half the time, move left half the time"

$$E[\text{"}left\text{"}] = E[U(now)] + E[U(later)] = 2 + 1 = 3 \tag{1}$$

$$E[\text{"}right\text{"}] = E[U(now)] + E[U(later)] = 2 + 3 = 5 \tag{2}$$

$$E[\text{"}1/2left + 1/2right\text{"}] = 2 + 1/2 + 3/2 = 4 \tag{3}$$

We pick here the policy that gives the highest expected value, and say that this is the value of getting dropped in this world in the middle. We have uncovered the policy function and determined the (maximum) value of the state.

2.2 General Steps to DP model

When solving these types of problems, you should follow these steps:

Step 1 Substitute constraint for c

Step 2 Solve for policy function

Step 3 Plug policy function back into Bellman

Step 4 Match up coefficients

3 Two quick Log Linearization examples

Follow these steps to log linearize:

Step 1 Take derivative of each side and evaluate at steady state

Step 2 Multiply by steady state variable

Step 3 Multiply by log-linearized variable

Step 4 Continue to next variable

Step 5 Simplify

3.1 From class

$$y_t = k_t^{\alpha} l_t^{1-\alpha}$$

$$1 * y * \hat{y}_t = \alpha k^{\alpha-1} l^{1-\alpha} * k * \hat{k_t} + (1-\alpha) k^{\alpha} l^{-\alpha} * l * \hat{l_t}$$

knowing that $y = k^{\alpha} l^{1-\alpha}$ we have

$$k^{\alpha} l^{1-\alpha} * \hat{y}_t = \alpha k^{\alpha} l^{1-\alpha} * \hat{k_t} + (1-\alpha) k^{\alpha} l^{1-\alpha} * \hat{l_t}$$

$$\hat{y}_t = \alpha \hat{k_t} + (1-\alpha) \hat{l_t}$$

3.2 From old comps

$$c_t n_t = (1-\tau_t) w_t$$

$$n * c * \hat{c}_t + c * n * \hat{n}_t = (1-\tau) * w * \hat{w}_t + -w * \tau * \hat{\tau}_t$$

$$cn(\hat{c}_t + \hat{n}_t) = (1-\tau) w \hat{w}_t - w\tau\hat{\tau}_t$$

$$\hat{c}_t + \hat{n}_t = \hat{w}_t - \frac{\tau}{1-\tau}\hat{\tau}_t$$

4 Solving the model

4.1 Conditions for a unique bounded solution

- $n_u = n_y$
- $|\lambda_i| \neq 1 \forall i$
- $R_{uy} \neq 0$ is invertible

4.2 Solving unstable and stable conditions

Two important equations are

$$\begin{bmatrix} s_{t+1} \\ u_{t+1} \end{bmatrix} = J \begin{bmatrix} s_t \\ u_t \end{bmatrix} + P^{-1} A^{-1} C \hat{z}_t \tag{4}$$

and

$$\begin{bmatrix} s_t \\ u_t \end{bmatrix} = P^{-1} \begin{bmatrix} \hat{k}_t \\ \hat{c}_t \end{bmatrix} \tag{5}$$

*Note here that if the order of $x_t = \begin{bmatrix} \hat{k}_t \\ \hat{c}_t \end{bmatrix}$ is switched then this would switch the k_t and c_t, and if the J matrix is stacked unstable then stable, then the s_t and u_t would switch as well. This would also change the R_{uy} portion of the P^{-1} matrix.

Then, to solve this part need to follow these steps:

Step 1: Set up equation to solve for u_t

Step 2: Iterate forward one period

Step 3: Substitute u_{t+1}

Step 4: Find that the part with u_{t+j} drops to 0 since it is multiplied by a number less than 1 raised to an infinite exponent

Step 5: Factor out the exogenous part and substitute law of motion for exogenous shock

Step 6: Set up infinite sum and remember that $\sum_{t=0}^{\infty} \beta^t = \frac{1}{1-\beta}$

Step 7: Plug the result for u_t

5 Steps for Solving Endogenous Growth Problems

The overall goal here is to get the growth rates in terms of constants, and that the growth rates of output, consumption, capital should be the same, and wages should be constant (no growth). Also, the constraints on the parameters should be that $\gamma_c > 1$. So whenever you get the growth rate in terms of constants, plug them back into expression for growth rate and say it needs to be greater than 1.

1. Set up HH lagrangian
2. Take FOCs
3. Solve for EE (γ_c)
4. [1]Use other equations and definitions from problem to solve for growth rate. Then set the EE equal to the growth rate found from definitions in problem to solve for constant ratio of endogenous variables, or, if not applicable then you should get the growth rate in terms of parameters.
5. Use firm's production function to argue why $\gamma_y = \gamma_k$
6. Say that by the BC and TVC $\gamma_c = \gamma_y = \gamma_k$.

6 Algorithm for Value Function Iteration

:

- Choose an initial guess V_0
- Set iteration counter $j = 0$ and tolerance $\epsilon > 0$
- Main step: on iteration j given values of V_j update V_{j+1}

$$V_{j+1}(k) = \max_{c,k'} u(c) + \beta V_j(k')$$

$$\text{s.t. } c + k' = f(k) + (1-\delta)k$$

- Continue iterations on the main step until $|V_{j+1} - V_j| < \epsilon$

[1]Note here that solving these problems sometimes is context specific. If you see something like $A_t = Y_{t-1}^{1/2}$ pretty good chance you should write out the definition of γ_A, and this should help you solve for the ratio of endogenous variables (think z/k or A/k).

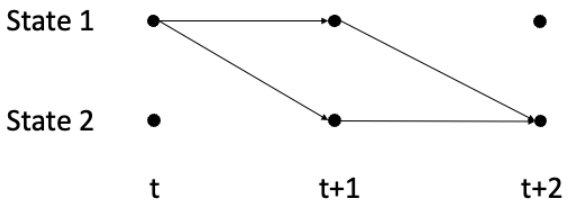
7 Transition matrices

Take the following transition matrix

$$\begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}$$

Compute the following

$$Prob(e_{t+2} = e_2 | e_t = e_1)$$



Where P_{ij} is the probability from moving from point i to point j

$$Prob(e_{t+2} = e_2 | e_t = e_1) = 0.8 * 0.2 + 0.8 * 0.6 = 0.64$$

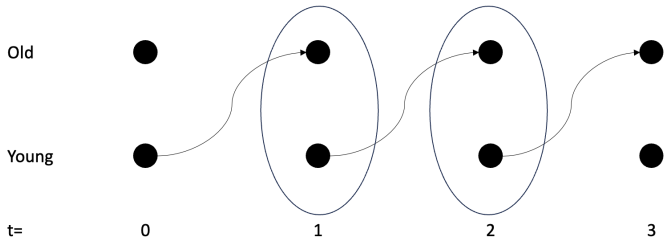
Define and solve for the stationary distribution π of the Markov chain. Stationary if $\pi_{t+1} = \pi_t$. This implies $\pi = \pi P$ and so $\pi(I - P) = 0$

$$0.8\pi_1 + 0.4\pi_2 = \pi_1$$

$$0.2\pi_1 + 0.6\pi_2 = \pi_2$$

Find they are colinear so add a normalizing $\pi_1 + \pi_2 = 1$ and $\pi_1 = 2/3$, $\pi_2 = 1/3$

8 OLG



In an OLG the strategy is to set up the budget constraints for the young and old, and then plug into the utility function and solve for the optimal savings rate s_t via an unconstrained optimization problem.

9 Dynamically Inefficient

The economy is dynamically inefficient if

$$k_{ss} > k_{GR}$$

where

$$k_{GR} = \arg\max [c = f(k) - (n + \delta)k]$$

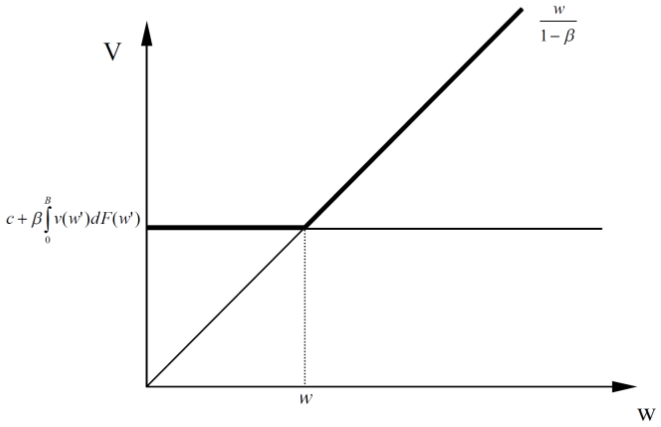
10 Search and unemployment

The employed problem is

$$V(w, e) = w + \beta \max_e \{V(w, e), V(w', u)\}$$

Plotting...

Since $V(w, u)$ is independent of w and $V(w, e)$ is increasing with w.



If the worker is employed

$$V(w, e) = w + \beta V(w, e)$$

Solve to find

$$V(w, e) = \frac{w}{1 - \beta}$$

The unemployment worker has benefits plus an uncertain wage in the future

$$V(w, u) = b + \beta \int_0^B V(w')dF(w')$$

10.1 Bathtub model of unemployment

A common way to analyze stock and flow problems in chemical engineering is the following general formula:

$$\text{accumulation} = \text{in} - \text{out} + \text{generation} - \text{consumption}$$

Here, if you prefer to not think of people being generated and consumed, you can reduce the equation to

$$\text{accumulation} = \text{in} - \text{out}$$

Looking at which people are entering into unemployment, that would be those who were employed and separated sE_t and the new cohort of workers entering into the labor force L_{t+1}. The people who are leaving unemployment are those who are hired from being unemployed fU_t and the old cohort who initially started in unemployment but got jobs L_t.

This reduces to

$$U_{t+1} - U_t = sE_t + L_{t+1} - fU_t - L_t$$

Reminder that the units here are people, not fraction of people unemployed. From here you can solve for the unemployment rate and steady state unemployment.